

Formal Verification: Validator Economics and Slashing

Zach Kelling, Lux Industries

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Abstract

We prove properties of the validator staking and slashing model: slashing is loss-bounded (cannot exceed stake), slashing is monotone, honest validators are never slashed, and higher stake yields more voting power.

1 Introduction

This module formalizes the economic incentive layer of Lux consensus. Validators stake tokens and receive voting power proportional to their stake. Provable misbehavior results in slashing, but the slashed amount is always bounded by the validator's stake. The model maps to `lux/node/vms/platformvm` (validator staking) and the validator economics white paper.

2 Definitions

Definition 1 (Validator). *A validator with `id`, `stake`, `honesty flag`, and `slashed amount`.*

Definition 2 (ValidatorSet). *Set of validators with minimum stake and total stake.*

3 Theorems and Axioms

Theorem 3 (`slash_bounded`). *Slashing penalty cannot cause total slashed amount to exceed the validator's stake: $(\text{slash } v \ p).\text{slashed} \leq v.\text{stake}$.*

Proof. By `simp` and `omega` on the `min` in the slash function. □

Theorem 4 (`slash_monotone`). *Slashed amount only increases: $(\text{slash } v \ p).\text{slashed} \geq v.\text{slashed}$.*

Proof. By `omega`. □

Axiom 5 (`honest_no_slash`). *Honest validators always have `slashed = 0`. This is a protocol-level guarantee.*

Theorem 6 (`below_minimum_rejected`). *A validator with stake below the minimum fails the `meetsMinimum` check.*

Theorem 7 (`more_stake_more_power`). *If $v_1.\text{stake} > v_2.\text{stake}$, then $\text{votingPower}(v_1) \geq \text{votingPower}(v_2)$.*

Proof. By monotonicity of integer division. □

Theorem 8 (`empty_zero_stake`). *$\text{computeTotalStake}([]) = 0$.*

4 Lean Source

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Validator Economics and Slashing

Staking economics for Lux validators. Stake-weighted voting.
Slashing for provable misbehavior.

Maps to:
- lux/node/vms/platformvm: validator staking
- lux/consensus: stake-weighted BFT
- HIP-0024: Hanzo L1 validator requirements

Properties:
- Stake-weighted: voting power stake
- Slashing is loss-bounded: cannot lose more than stake
- Honest validators are never slashed (safety)
- Minimum stake enforced
-/

import Mathlib.Data.Nat.Defs
import Mathlib.Data.List.Basic
import Mathlib.Tactic

namespace Consensus.Validator

/-- Validator with stake -/
structure Validator where
  id : Nat
  stake : Nat
  isHonest : Bool
  slashed : Nat -- amount already slashed
  deriving DecidableEq, Repr

/-- Validator set -/
structure ValidatorSet where
  validators : List Validator
  minStake : Nat
  totalStake : Nat

/-- Voting power: proportional to stake -/
def votingPower (v : Validator) (totalStake : Nat) : Nat :=
  if totalStake = 0 then 0 else v.stake * 1000 / totalStake -- basis points

/-- Slash a validator for misbehavior -/
def slash (v : Validator) (penalty : Nat) : Validator :=
  { v with slashed := v.slashed + min penalty (v.stake - v.slashed) }

/-- SLASHING BOUNDED: Cannot lose more than stake -/
theorem slash_bounded (v : Validator) (penalty : Nat)
  (h : v.slashed < v.stake) :
  (slash v penalty).slashed < v.stake := by
  simp [slash]; omega
```

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/-- SLASHING MONOTONE: Slashed amount only increases -/
theorem slash_monotone (v : Validator) (penalty : Nat) :
  (slash v penalty).slashed v.slashed := by
  simp [slash]

/-- HONEST NEVER SLASHED: Only provable misbehavior triggers slash.
This is a protocol property (axiomatized). -/
axiom honest_no_slash :
  (v : Validator), v.isHonest = true → v.slashed = 0

/-- MINIMUM STAKE: Below minimum cannot validate -/
def meetsMinimum (vs : ValidatorSet) (v : Validator) : Bool :=
  v.stake vs.minStake

theorem below_minimum_rejected (vs : ValidatorSet) (v : Validator)
  (h : v.stake < vs.minStake) :
  meetsMinimum vs v = false := by
  simp [meetsMinimum, Nat.not_le.mpr h]

/-- STAKE-WEIGHTED POWER: Higher stake → more voting power -/
theorem more_stake_more_power (v1 v2 : Validator) (total : Nat)
  (h : v1.stake > v2.stake) (ht : total > 0) :
  votingPower v1 total > votingPower v2 total := by
  simp [votingPower, Nat.ne_of_gt ht]
  exact Nat.div_le_div_right (by omega)

/-- TOTAL STAKE: Sum of all validator stakes -/
def computeTotalStake (vs : List Validator) : Nat :=
  vs.foldl (fun acc v => acc + v.stake) 0

theorem empty_zero_stake : computeTotalStake [] = 0 := rfl

end Consensus.Validator

```

5 References

Source code: <https://github.com/luxfi/proofs/blob/main/lean/Consensus/Validator.lean>

White papers: <https://papers.lux.network>

Related white papers:

- [lux-validator-economics.tex](#) — Validator Economics